

CompTIA A+ AP-A1

Hardware Components & System Assembly

Easy English Notes

Purpose: Simple notes for the assessment topics and practical tasks.

1. Selecting PC Components

Before building a PC, make sure the motherboard, CPU, RAM, storage and PSU are compatible.

Motherboard: Choose the correct CPU socket, RAM type, storage support and required ports.

CPU: Check that the CPU socket and chipset support the processor.

RAM: Check the RAM type, speed support and maximum capacity of the motherboard.

Storage: Check whether the motherboard supports SATA drives and/or M.2 NVMe drives.

Compatibility check: Use the motherboard and CPU specifications to confirm that all parts work together.

2. HDD, SSD and NVMe

Storage	Main Point	Best Use
HDD	Uses moving parts; slower and cheaper.	Large low-cost storage.
SSD	Uses flash memory; faster and quieter than HDD.	Office and everyday PCs.
NVMe SSD	Uses PCIe and is usually faster than SATA SSD.	Fast boot, apps and heavy work.

For a modern office workstation, an SSD is a good choice because it gives faster startup and file access than an HDD.

3. GPU and Cooling

GPU: The Graphics Processing Unit handles graphics and visual work. It helps graphics-intensive applications run smoothly.

Cooling: Cooling removes heat from the CPU and other components. Good cooling helps prevent overheating and shutdowns.

4. Hardware Compatibility

- The CPU must match the motherboard socket.
- RAM must be supported by the motherboard.
- Storage must use a supported connection such as SATA or M.2.
- The PSU must provide enough power and the correct connectors.
- Wrong or incompatible parts may cause no boot, crashes, or hardware not being detected.

5. PSU Selection and Safety

Choose a PSU with enough wattage for the CPU, GPU and other components, with some extra capacity.

- Turn off and unplug the PC before making connections.
- Use the correct motherboard power connector.
- Connect CPU power near the CPU socket.
- Connect GPU power when required.
- Connect SATA power to SATA drives.
- Do not force connectors into the wrong socket.
- Check all connections before the first power-on.

6. POST and Boot Troubleshooting

POST means Power-On Self-Test. It is the hardware check performed when the PC starts.

Successful POST: The PC completes the startup hardware check and continues to boot.

If POST fails: Check power, motherboard connections, RAM, CPU, GPU and storage in a logical order.

7. Core Hardware Components

Component	Simple Function
Motherboard	Connects and allows communication between PC components.
CPU	Processes instructions and performs calculations.
RAM	Temporarily stores data used by running programs.
GPU	Processes graphics and video output.
Storage	Stores the operating system, programs and files.
Cooling System	Removes heat and keeps components at safe temperatures.

8. Motherboard Ports and Slots

- SATA: connects SATA storage devices.
- M.2: supports small internal drives such as NVMe SSDs.
- PCIe: used for expansion cards such as graphics cards.
- USB headers: connect front-panel USB ports.
- Power connectors: provide power to the motherboard and CPU.
- CPU socket: holds the processor and must match the CPU type.

9. RAM and Storage Installation

- Turn off the PC and disconnect power.
- Use ESD safety precautions.
- Open the RAM slot clips and remove the RAM carefully.
- Align the RAM notch and press it down until the clips lock.
- Disconnect and reconnect the storage device using the correct SATA or M.2 connection.
- Power on and check that RAM and storage are detected in BIOS/UEFI.

10. Full Desktop PC Assembly

- Prepare the motherboard, CPU, RAM, storage, GPU, PSU and case.
- Install the CPU and cooler correctly.
- Install RAM and storage.
- Install the motherboard in the case.
- Connect motherboard, CPU and storage power.
- Install the GPU if required and connect its power.
- Connect front-panel cables.
- Check all connections.

- Power on and confirm successful POST.

11. Boot Failure Diagnosis

If a newly assembled PC does not boot, start with simple physical checks before software checks.

Priority	Check	Why
1	PSU and power cables	No power or loose power connections can stop the PC.
2	RAM	Loose or faulty RAM can prevent POST.
3	Motherboard/CPU	Wrong installation or compatibility can stop startup.
4	GPU	A faulty or badly seated GPU can cause no display.
5	Storage	A missing boot drive can prevent the operating system from starting.

12. Installation Verification Checklist

- Confirm every component is compatible.
- Confirm RAM is fully seated and locked.
- Confirm the storage device is connected correctly.
- Confirm CPU and motherboard power connectors are secure.
- Confirm GPU and GPU power are connected if needed.
- Power on and check POST.
- Check BIOS/UEFI to confirm RAM and storage are detected.

13. Why Use SSD Instead of HDD?

SSD is suitable for modern office workstations because it is faster, has no moving parts, works quietly and gives quick system startup and file access. An HDD can still be useful when large storage capacity at low cost is the main requirement.

14. Simple Junior Technician Boot Checklist

1. Confirm the power cable and PSU switch.
2. Check motherboard and CPU power connectors.
3. Check RAM seating and test one module at a time if needed.
4. Check GPU seating and power if there is no display.
5. Check whether storage is detected in BIOS/UEFI.
6. Check CPU and motherboard compatibility.
7. Check POST lights or beep codes.
8. Correct the physical problem and test boot again.

15. Practical Safety Rules

- Always shut down and unplug the PC before installing or removing hardware.
- Use ESD protection when handling components.
- Hold RAM, GPUs and other boards by their edges.
- Do not force connectors.
- Keep screws and tools away from the motherboard.

- Check every connection before powering on.

Quick Revision

CPU = Processing • **RAM** = Temporary memory • **GPU** = Graphics • **SSD/NVMe** = Fast storage • **HDD** = Low-cost large storage • **Motherboard** = Connects components • **PSU** = Supplies power • **Cooling** = Removes heat • **POST** = Startup hardware test.